

CS 347: Software Engineering

Course Code: CS 347

Course Name: Software Engineering

L-T-P-C: 3-0-2-4

Prerequisites: None

Intended for: B.tech CSE (Third Year)

Mutual Exclusion: None

Approval: 56th BoA; 65th BoA (Second Major in CSE list: replaced by CS 303 Software Engineering, 2-0-2-3)

Preamble

The course introduces students to the different phases of Software Development Life Cycle (SDLC) used in maintaining, developing and delivering software products. Student will acquire basic software development skills and understand common terminology used in software engineering profession.

Course Modules

- **Unit 1: Introduction:** software engineering principles, software life-cycle models, software requirements specification, formal requirements specification and verification - axiomatic and algebraic specifications (9 Hours)
- **Unit 2: Software Architecture and Design:** Function-Oriented Software design, Object Oriented Design, UML, design patterns, user interface design, computer-aided software engineering (CASE), software reuse, component-based software development, extreme programming. (12 Hours)
- **Unit 3: Software Testing and Verification:** Coding and Unit testing, Integration and System testing, debugging techniques, Software Quality: SEI CMM and ISO-9001, Software Reliability and fault tolerance. (8 Hours)
- **Unit 4: Software Management:** Evolution, Project Management and Risk Analysis, Software Quality Management, Configuration Management, Software Metrics, Cost Analysis and Estimation, Manpower Management and Organization and Management of large software projects. (11 Hours)

Laboratory/Practical/Tutorial Modules

Lab(s)/Assignment(s) related to different aspect of software development, testing and verification using CASE tool and workbenches.

Text Books

1. Pressman, *Software Engineering: A Practitioner's Approach*, McGraw Hill Education, 7th edition, ISBN: 9780071267823

References

1. I. Sommerville, *Software Engineering*, 5/e, Addison-Wesley, 2000.



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2. T. C. Lethbridge and Robert Laganire, *Object Oriented Software Engineering*, Tata McGraw Hill, 2004.
 3. Bennett, S. McRobb, R. Farmer, *Object Oriented Systems Analysis and Design using UML*, McGraw-Hill, 4th edition, 2010, ISBN 9780077125363